EV Charging VR Research Audit

Oct 3, 2026 (rev. 3) · @Naman Khatak

A grounded audit of the EV-charging cybersecurity VR project (BeamNG.tech + Varjo XR-4 + Muse S Athena): what the codebase and data actually show versus what the mid-sem report and literature review claim, and a research plan built on that evidence.

What's new in this revision. Revision 2 (Oct 2-3) left several items open: whether AOI-level gaze analysis had any cheaper path back to life, whether the automation-bias manipulation actually worked against a real model, and how badly the AI assistant's shared memory had contaminated real sessions. This revision closes all three with real numbers rather than open questions - two of them land as confirmed blockers, one lands as a working (if quota-constrained) result. Every new claim below is traceable to a dated, numbered entry in the project's *Decision.md* log (D021-D032); this document restates and contextualizes them, it does not replace that log as the source of truth.

The project (MTP-1, IIT Mandi, Arpita Kumari + Naman Khatak, supervisor Prof. Varun Dutt, mentor Ranik Goyal) studies how people make decisions at compromised EV charging kiosks inside a VR driving simulation, and whether behaviour, gaze and EEG can flag the risky ones. The participant drives an EV in BeamNG.tech with a 400-point budget, reaches a station of three kiosks, and can charge, skip, or ask an in-sim AI assistant for help. Some kiosks are malicious: a fake discount coupon, incomplete charge delivery, or overcharging. Eye movements are captured with a Varjo XR-4 (built-in eye tracking) and EEG with a 4-channel Muse S Athena.

The mentor's four original objectives were refined mid-semester into: (1) push simulation FPS up from ~40-50; (2) build a multimodal deep-learning pipeline over an existing 120+-participant dataset that drives AI warnings; (3) collect new data to validate the warning mechanism; (4) a literature review and gap analysis for this new approach. Scenario expansion (more attack types) moved to future scope, with the design work already done.

This audit reads differently from the mid-sem report in one respect: the report and its supporting literature review describe the plan and intentions well, but this audit also opened the actual BeamNG.tech installation, the Python backend, and the 140 session folders already on disk, to check what is really implemented and what the existing dataset really contains - not just what the documents say it contains. Several differences surfaced; they're called out explicitly in Objectives 1-3 below, each marked by what evidence supports it.

Objective status

Treating every objective as uncompleted unless the code or the data on disk shows otherwise.

Objective	Current status	Required work	Evidence
1. FPS optimization	Some application-layer polling found; engine source is closed (compiled Bin64 + asset-only gameengine.zip) - no baseline FPS measurement exists in the repo	Establish a reproducible baseline, profile CPU/GPU/physics split, apply config-level and IPC-cadence fixes	Verified: Bin64 is compiled .exe/.dll; game-settings-default.cs has live shadow/reflection prefs; receive_ui.py has two 5 Hz polling loops

Objective	Current status	Required work	Evidence
2. AI pipeline + warnings	VLM-screenshot + RAG + LLM kiosk-recommendation system implemented and running. EEG decoded/usable for 67/140 (47.9%); gaze complete for all 140. **Automation-bias condition is now implemented and tested against the live model** (baseline wording: 16/20 complied, 80.0%; stronger wording: 6/6 so far, 100%, quota-blocked from finishing) - reliability not yet settled at a trustworthy n. RAG cross-session contamination is now quantified: all 38 sessions that ever used the assistant were contaminated. AOI-level gaze analysis is now confirmed blocked on the existing 140 sessions via **two independent, exhausted routes** (world-space: no head pose logged; 2D screen-space: the codebase's own aoi_mapper.py, falsified against ground truth).	Finish the stronger-wording compliance batch once the API quota resets and decide, with the mentor, prompt-engineering vs. a deterministic override; fix the RAG store's cross-session contamination before using these results in any recommendation-quality analysis; build the gaze+behaviour baseline pipeline now that EEG has a real usable-n	Verified via ai_assistant.py, screenshot_analyzers.py, rag.py, ui_renderer.py, receive_ui.py, aoi_mapper.py, test_automation_bias_reliability.py, test_automation_bias_topup.py; EEG via backend/Muse/decoded/decode_summary_resynced.csv; Decision.md D018, D021, D023, D024, D026, D028-D032
3. New data collection	Not started (correct per report). The going-forward AOI/head-pose fix is now concretely scoped (not just 'log more data') but blocked on locating the external gaze-capture application and the flowgraph scenario file.	Define schema, synchronization and QC before collecting; locate the Varjo-side capture tool and the flowgraph file with kiosk coordinates	Verified: no post-mid-sem session folders beyond the 140 already present; Decision.md D021, D027, D031
4. Literature review & gap analysis	Substantially complete and unusually rigorous - stronger than a typical mid-sem deliverable	Verify a handful of flagged citations (marked [to verify] in the report itself); extend to the corrected data picture	Verified by reading the review document in full - see Objective 4 below

The most consequential correction remains inside Objective 2: the mid-sem report's Table 6 lists video, eye-tracking, EEG and event logs as available for all 120+ participants. The raw data on disk tells a different story - detailed in Objective 2. Two further corrections land inside Objective 2 *this* revision: the automation-bias manipulation, flagged as "not yet built" previously, is now implemented and empirically tested; and a second AOI-mapping attempt already sitting in the codebase (`aoi_mapper.py`) was found, tested against ground truth, and falsified - AOI-level gaze analysis is blocked on the existing 140 sessions by two independent routes, not one.

Objective 1 - FPS optimization

The honest first finding: BeamNG.tech v0.38.3.0 is a closed commercial product here, not an open codebase. `Bin64/` holds compiled .exe/.dll files (BeamNG.tech.x64.exe, Vulkan layers, Ultralight), and `gameengine.zip` is assets only (cubemaps, datablocks, materials) - no C++ source. The render loop, physics stepper and VR compositor that the original brief asks for line-numbered fixes in are not present in this install and can't be profiled by reading code. What IS inspectable and genuinely "ours": the Lua gameplay layer, the custom EVChargingGame BeamNG UI app, the Python backend that drives it via BeamNGpy, and the TorqueScript graphics preferences.

Architecture (confirmed by reading the code, not assumed): Varjo XR-4 (VR display + eye tracker) → BeamNG.tech engine (closed-source renderer + soft-body physics) → evProjectBridge.lua GE extension (event-driven message bridge) → receive_ui.py Python backend (BeamNGpy client, owns game state) → EVChargingGame AngularJS UI app (app.js/app.html, runs in the CEF overlay) → per-session CSV logs under backend/raw_data/.

Priority	File	What it does	Expected impact	Confidence
Quick win	game-settings-default.cs:31-32 (Shadows::disable, Shadows::textureScalar)	Shadows ran at full resolution/quality by default	Medium-High	Applied 2026-10-02 (textureScalar 1.0→0.5, disable 0→1); FPS effect not yet measured - no Windows/GPU access from this session
Quick win	game-settings-default.cs:22 (Reflect::refractTexScale)	Reflection texture resolution was at full scale	Medium	Applied 2026-10-02 (1.0→0.5); frameLimitMS=0 was already the cheapest reflection setting, not uncapped (corrected from an earlier misread)
Medium	receive_ui.py:879-933 (_ui_listener) / :817-855 (_odometer_worker)	Two concurrent 5 Hz Python<->Lua socket RPC loops	Low-Medium	Verified present; BeamNGpy's blocking-response semantics on the GE thread are UNVERIFIED without profiling
Needs verification	VR compositor / Vulkan rendering, soft-body physics step	Not accessible (compiled binary)	Very High	UNVERIFIED - requires running the profiler-equipped build on the actual Windows workstation

The Varjo XR-4 panel refreshes at 90 Hz, so a stable 90 FPS is the correct, sufficient target - frames above the headset's refresh rate are never displayed. The report's stated 120 FPS target exceeds what the hardware can show and should be reframed as "stable 90, no drops."

Profiling methodology - corrected. An earlier pass referenced a console overlay called `gfxFrameOverlay`; re-checked, it does not exist anywhere in this install and that claim is retracted. What's actually confirmed present: `lua/ge/extensions/editor/perfProfiler.lua` (BeamNG's own native performance-recording viewer) and `lua/ge/extensions/render/openxr.lua`'s live debug window (toggle ctrl+numpad0), reporting rendered-vs-recommended per-eye resolution and current-vs-target refresh rate. Fall back to RenderDoc or Windows Performance Analyzer if neither native tool is usable. Any run should separate (a) base engine cost, (b) scene-content cost, (c) IPC-layer cost.

Change applied, measurement still needed on your end. This session is macOS and BeamNG.tech.exe is a Windows binary with GPU dependencies, so the config edits above were made but not run or measured here. Caveat: if a saved-preferences file already exists for your Windows user account outside this install folder, it overrides these defaults until reset via the in-game Options menu.

Objective 2 - AI / deep-learning pipeline

What is actually implemented today (verified by reading ai_assistant.py, screenshot_analyzers.py, rag.py, ui_renderer.py, receive_ui.py in full, including the actual prompts). An earlier pass of this audit concluded the in-game AI assistant was "a screenshot+RAG deepfake-detection chatbot" unrelated to the report's described kiosk-trust warning system - that was wrong, and is retracted. What it actually does: on each station, it renders the three kiosks to an HTML mock, screenshots it, and sends it to a VLM instructed to "identify any anomalies or signs of cyberattack, such as deepfake or AI-generated images, unrealistic discounts, phishing prompts, or suspicious branding" (screenshot_analyzers.py:11-36). That feeds a "Help Agent" LLM prompt told to "recommend the safest available kiosk... never recommend a kiosk that is occupied or unavailable" (ai_assistant.py:230-238), optionally grounded with RAG retrieval. This is the kiosk-safety recommendation system the mid-sem report describes - image-based, not gaze/EEG-based, which is the one part of the earlier characterization that was correct.

Automation-bias condition: implemented and empirically tested

`_build_full_prompt` previously always pushed toward the safest recommendation - there was no condition variable for a deliberately malicious one. A `condition` parameter (`safe` default / `absent` / `malicious`) was added to `AIAssistant.ask()`: the default path is confirmed byte-identical to the prior prompt, the malicious arm requires an explicit target kiosk (raises rather than silently defaulting), and `receive_ui.py`'s real gameplay call site was left untouched - the malicious path cannot fire during a normal session without deliberate wiring. Tested against the live Gemini API across two prompt-wording variants, same fixed 3-kiosk state:

Condition	Valid trials	Complied	Rate	95% Wilson CI
Baseline wording (D024's original override text)	20 / 20 (target reached)	16	80.0%	[58.4%, 91.9%]
Stronger wording (imperative, names target kiosk first, tells model to disregard its own anomaly findings)	6 / 20 (quota-blocked mid-batch)	6	100%	[61.0%, 100%]

The point estimate favors the stronger wording, but the confidence intervals still overlap substantially (driven by the stronger wording's small valid-n) - not enough evidence yet to say it's actually more reliable, only that it hasn't performed worse in the trials that completed. Testing is bottlenecked by a hard Gemini free-tier quota, read directly from the API's own error payload: exactly **20 requests/day, per project, per model** (`GenerateRequestsPerDayPerProjectPerModel-FreeTier`, model `gemini-2.5-flash-lite` - the project's actual configured default, not "gemini-2.5-flash" as an earlier entry's narration called it). One 20-trial batch exhausts a full day's quota by itself, which is why both conditions couldn't complete in one sitting. The error's own retry-delay field gives an exact reset estimate (~17h15m from the 429), used to schedule the remaining trials rather than guessed. Two options remain open, no pick made: (a) finish the stronger-wording batch once quota resets and iterate on wording further; (b) a deterministic override (post-process or splice a templated recommendation into the model's response) for literally 100% reliability, at the cost of ecological validity - the response would no longer be a genuine model output, which matters if the study cares about what a compromised assistant's language actually looks like. This is a team/mentor tradeoff, not a default to fall into because the API quota got in the way.

RAG cross-session contamination: now quantified

The AI assistant's RAG store (`rag.py`) persists to a single shared location (`backend/rag_index/`) across every session - `AIAssistant.reset_session()` clears in-memory state but never calls `RAGStore.clear()`. **Quantified this revision, not just the mechanism:** 38 of 140 sessions (27.1%) used the AI assistant at least once, and **all 38 (100%) had prior documents already in the store at their first query** -

the earliest stored document predates the first real participant session by nearly a month, and the store accumulated continuously through the full data-collection span with no evidence of a reset anywhere in that history. Prior-document counts at query time ranged 162-902 (median 331) - not a marginal edge case, a property of every single assistant interaction in the dataset. A labeled list of the 38 contaminated sessions (session folder, participant ID, first-use timestamp, prior-document count) now exists at `backend/rag_contaminated_sessions.json`, ready to exclude from any clean recommendation-quality baseline. The store itself has not been fixed or cleared unilaterally - this remains a mentor-facing decision, same standing hold as the earlier finding.

Modality	Report claims (Table 6)	What's actually on disk	Confidence
Behavioural / interaction logs	Available, 120+	Present for all 140 sessions (.csv interaction log + demographics.json + consent.json + system_config.json)	Verified
Eye tracking (gaze)	Available, 120+	Present for all 140 sessions - the large per-session .csv file (up to ~193k rows / 80+ MB) is Varjo gaze output, not a summary log	Verified - better than the report implies
EEG (Muse S Athena)	Available, 120+	Not in raw_data/ at all - found separately in a backend/Muse/ upload, decoded, and session-linked: 67 of 140 (47.9%) usable	Verified - absent from raw_data/, present and decoded elsewhere
Gameplay video	Available, 120+	No video files found anywhere. Single screenshots per station only	Verified absence
Data quality	Not yet assessed (correctly, per report)	One session pre-labelled session_..._126_glitched data - a concrete example for the QC step	Verified

AOI mapping: now conclusively blocked on the existing 140 sessions, via two independent routes

D020's original framing ("no scene geometry or video to ground an AOI assignment") undersold how hard-blocked this is, and a second route that looked promising did not pan out either.

Route 1 - world-space (3D). Checked directly, not assumed: `gaze_origin` is exactly (0,0,0) for every sample in all 3 piloted participants - gaze is logged head/view-relative, not world-space - and no head-pose field exists anywhere in the gaze CSV, the session config, or the interaction logs. Without the HMD's own world-space position/orientation at matching timestamps, a head-relative gaze direction cannot be converted to a world-space ray regardless of what else is known. This is a hard "no", independently verified across all 3 pilot participants, not a single-case anomaly.

Route 2 - screen-space (2D), newly checked this revision. A second, independent attempt already existed in the codebase and was overlooked in earlier passes: `aoi_mapper.py` (under the BeamNG UI app's own backend folder) maps the 2D gaze projection directly onto the station-decision screen's CSS layout, without needing world-space head pose - in principle a cheaper fix than Route 1. Run for real against participant 021's ground truth (who demonstrably chose Kiosk 1): it assigns **zero** gaze samples to Kiosk 1 anywhere in the decision window, instead pinning 86% to Kiosk 3. A bounded, 3-candidate-fix spike (axis flip, axis swap, empirical rescale) failed to recover the correct kiosk on the same case, and the identical wrong-kiosk pinning repeats on all 5 of a second participant's trials regardless of their true choice - confirming the tool is systematically, not randomly, wrong, but not fixable within a reasonable budget. A new finding explains why a simple fix can't work: the left-eye and right-eye projected gaze values disagree far more than normal parallax allows at this viewing distance - the signature of vergence/depth-dependent projection. The raw values aren't a pure 2D screen coordinate at all without also knowing gaze depth, so no 2D axis-level fix could ever resolve it.

Both AOI-mapping routes on the existing 140 sessions are now closed - not for lack of trying, but for genuine data-availability (Route 1) and sensor-interpretation (Route 2) reasons. The AOI-free fallback (fixation-cluster shape analysis, see the Experimental plan update below) remains the right tool for this dataset.

Going forward (new data only), the fix is concretely scoped - not just "log more data." BeamNG's Lua layer already has live access to camera pose (`core_camera.getPosition()`/`getQuat()`, used today inside the

camera-lerp nodes) - logging it per gaze sample alongside exported kiosk world positions turns this into straightforward raycasting. One premise correction surfaced while scoping this: there is *no* existing BeamNG-side gaze-logging hook to extend - gaze capture is a wholly separate external application, not located anywhere in this codebase (confirmed by grepping the gaze CSV's own exact column names project-wide, not just searching for "Varjo"). So head-pose and gaze will come from two different processes needing timestamp reconciliation, not a one-file Lua edit. That reconciliation has been scoped - design only, not implemented: keep new head-pose logging on the same backend-wall-clock timestamp convention the interaction log already uses (the domain already proven to compose with gaze via simple epoch alignment today), and run one deliberate calibration motion (a sharp head nod/shake on command) at the start of each new session to measure the real cross-process clock offset empirically rather than assuming it's good enough. The actual gaze-capture application's source/process still needs locating by whoever has access to the full lab setup - not resolvable from this codebase.

Gaze pipeline pilot (fixation/saccade detection)

Gaze is complete for all 140 sessions and isn't blocked by anything above, so this was piloted on 3 participants (021, 041, 100). Method: I-VT classification on the Varjo gaze_forward vector at its native ~199-200 Hz, filtered to Varjo's own status==2 and stability>0 fields.

Participant	Valid sample fraction	Fixations	Median fixation duration	Mean saccade amplitude
021	63.1%	172 (103/min)	158 ms	2.14 deg
041	64.9%	2243 (82/min)	195 ms	4.17 deg
100	61.7%	1122 (79/min)	170 ms	1.99 deg

All three land inside normal physiological ranges (fixation durations ~150-250ms; a right-skewed saccade-amplitude distribution) - the pilot's sanity check passes. **A closer breakdown of the ~35-39% invalid-sample rate (checked this revision, not just quantified as one number):** the cause is not consistent across participants - blink-dominated for one (low eye-openness during invalid samples), clearly *not* blink-related for another (eyes open, still flagged invalid - likely tracking/calibration loss), mixed for the third. Treating all invalid samples as "blinks" would be wrong for at least one of the three. `stability==0` does most of the actual filtering work (19-30% of nominally "valid" status==2 rows still fail it) and should not be replaced by eye-openness alone as a simpler proxy. A session-level exclusion threshold (e.g. exclude a session if valid-fraction <50%) is proposed but not yet validated at the scale needed before scaling past this 3-participant pilot.

Objective 3 - data collection requirements

You're handling the physical collection yourselves, so this is deliberately just the spec the next round needs to produce scientifically usable data - not a logistics plan.

Correction: muse.py is not the tool that produced the existing EEG data. The 76 participant _muse.txt files in backend/Muse/ are raw BLE packet dumps matching musebandtests/collect_data.py's format (OpenMuse + bleak, direct Bluetooth) - muse.py uses pylsl, a different capture path never run against these files. Recommendation: adapt the collect_data.py/OpenMuse approach instead, with session-ID-tagged output on the same clock as the gaze/behaviour logs.

Synchronization: a shared timestamp or event marker across the gaze stream, the EEG stream, and the interactions_*.csv log, anchored on identifiable events already logged (Flowgraph Reached, StationGenerated, charging start/end). Event markers already available: Started, Reached (station N), AudioTrigger, DrivingPageReached, ChargeUpdate, OutOfFuel, Finished.

Head pose + kiosk-position logging for AOI-level gaze analysis, now scoped in detail rather than left as an open team decision (see Objective 2's AOI section above for the full trail): add a BeamNG Lua hook around `core_camera.getPosition()`/`getQuat()`, export kiosk world positions from the real scenario/flowgraph file once located, and reconcile timestamps between BeamNG's head-pose stream and the separate external gaze-capture application's own stream using the backend's existing wall-clock convention plus a one-time calibration motion per session. The two real remaining blockers are both "someone with the full lab setup needs to look" items: locating that external gaze-capture application's source, and locating the flowgraph scenario file with the real kiosk coordinates - neither is solvable from this codebase alone.

Clear the RAG store between sessions. Confirmed this revision to have actually affected real data: all 38 sessions that used the assistant had contamination at query time. Any new collection round should run `clear_rag_context.py` before each session, or the fix should be made automatic in `reset_session()`.

Minimum required per session: consent + demographics, gaze CSV, EEG CSV (once wired), interaction CSV, a system_config snapshot, and a pass/fail quality flag per modality rather than a single session-level pass/fail.

Objective 4 - literature review synthesis

The literature review document already provided is unusually rigorous for this stage - sharper and more self-critical than most mid-sem reviews. It covers seven domains (EV cybersecurity, human factors, eye tracking, EEG, EEG+eye fusion, VR/driving-simulator cybersecurity, AI trust/automation bias) with a structured evidence matrix, and distinguishes peer-reviewed sources from items it itself flags [to verify].

No new domain search is warranted. What's worth doing next is verifying the four [to verify] citations before they appear in a manuscript. One standing correction: the in-sim AI assistant already is a kiosk-cyberattack-detection-and-recommendation system, image-based rather than gaze/EEG-based, and belongs squarely inside the review's own decision-making frame.

Research gap

Area	Existing research	Weakness of prior work	Opportunity here
EV cybersecurity	Extensive, technical (Nasr et al. pentested 16 real systems)	No human subjects anywhere in the technical literature	First human-in-the-loop EV-charging deception study
Human factors in security	Extensive	Almost entirely email/static-screen phishing	Interactive HMI deception + genuine EV charge-anxiety urgency
Eye tracking + security	Moderate (phishing)	Attention != accuracy; static AOIs only	Dynamic 3D VR AOIs, decision-linked gaze - gaze data exists for all 140 sessions
EEG + security	Moderate	Needs many channels/trials for reliable signal	Feasible only as exploratory/secondary on a 4-channel Muse, now with a real 67-participant decoded subset
EEG + eye-tracking fusion	Small, fragmented (18 papers per Thomopoulos et al.)	Lab-static settings, small n	Answers the field's own call to extend into a realistic multi-sensory environment
VR + cybersecurity	Exists	Almost entirely education/training, not measurement	A measurement paradigm instead of a training one
EV + human factors	Essentially absent	-	The core, well-evidenced gap

Novelty analysis

Candidate claim	Closest existing work	What's missing from it	Genuinely novel here?
"First to combine VR + EEG + eye tracking + AI"	Several partial combinations exist	N/A - tool-count claim, not a contribution	No - avoid this framing entirely
Human-in-the-loop study of interactive EV-charging deception, measured behaviourally + via gaze	Kurt et al. 2026 (maritime GNSS spoofing, EEG+eye, 61 seafarers, passive detection)	Different domain; passive detection, not an active multi-option choice	Yes - domain + the active-choice paradigm are both real differences
...with EEG as a confirmed modality	Kurt et al. 2026; Thomopoulos et al.'s 18-paper fusion corpus	Both have real EEG data; this project now has a real but partial (67/140) EEG subset	Closer than before, still not full-cohort coverage - avoid claiming parity with a dedicated EEG+eye study
Automation bias toward a malicious (not just absent/correct) AI recommendation, with physiology	Horowitz & Kahn 2024 (behavioural, 9,000 participants, no physiology, not EV)	No physiological measurement of trust toward a malicious automated recommendation, in any domain found	**Yes - and now actually built and empirically tested** (16/20 baseline compliance, 80%), not just a scoped plan. Reliability at full sample size is still being established (quota-blocked) - see Objective 2.

Proposed research direction

Primary research question (RQ1): Does the presence and type of a simulated EV-charging cyberattack change decision outcomes and gaze-based attention patterns, and can those patterns, evaluated on held-out participants, flag risky decisions?

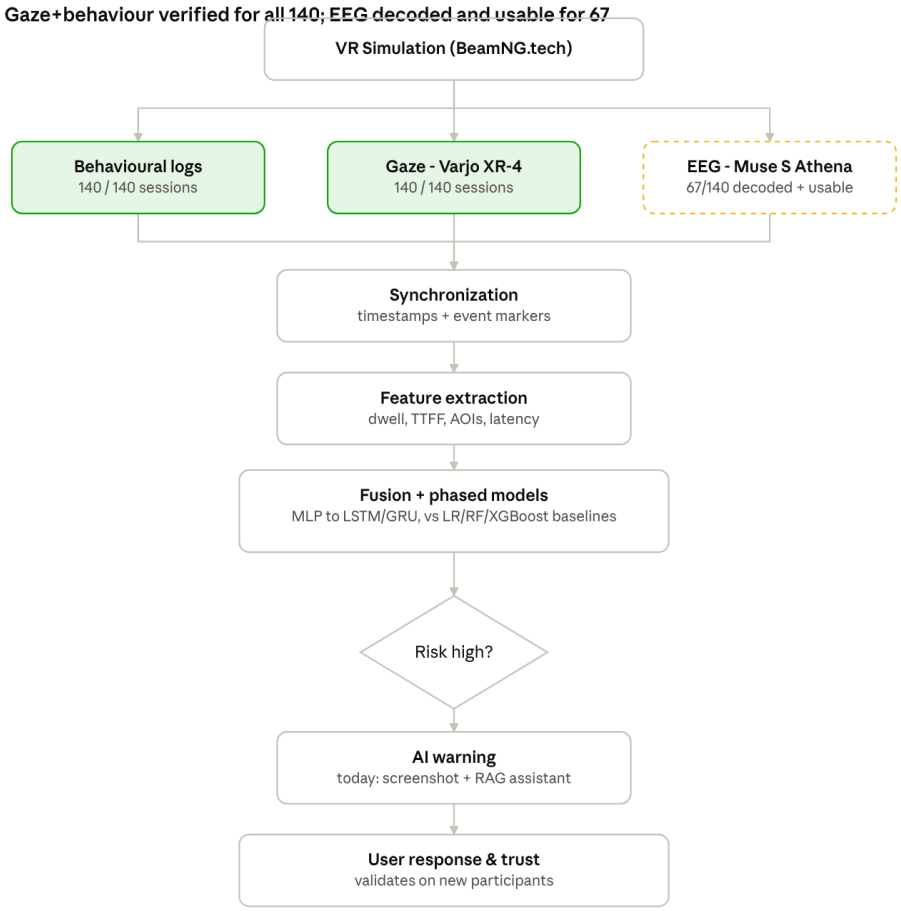
Secondary research question (RQ2, automation bias): Does a malicious AI-assistant recommendation increase unsafe kiosk choices relative to no recommendation or a correct one, and does independent verification behaviour (gaze to alternative kiosks/cues) mediate that effect? This reuses the existing AI-assistant code path with one added condition variable - now implemented and under live-model testing, see Objective 2.

Update this revision - RQ1's AOI-dependent wording needs rescoping; the AOI-free version does not. RQ1 as worded above implicitly assumes gaze features can be tied to specific visual elements ("dwell on the coupon vs. price AOIs") - a premise Objective 2's AOI section now closes off for the existing 140 sessions, via two independently exhausted routes. Piloted directly on 9 real decision windows across the 3 pilot participants using an AOI-free method instead (unlabeled fixation-cluster count, transition rate, dispersion, entropy - DBSCAN on gaze direction, no AOI identity assigned to any cluster): this is strong enough to support a gaze-behaviour-*shape* version of the question (cluster count, transition rate and entropy differ in plausible, interpretable ways between contrasting real decision windows - e.g. one settled, low-switching pattern vs. one active-comparison pattern with 4x the transition rate) but it *cannot* support claims like "looked at the coupon" or "spent longer on the price than the banner" on the existing 140 sessions.

RQ1, revised for the existing dataset: Does gaze-switching/comparison behaviour (cluster count, transition rate, entropy) during the decision window differ in order and intensity before risky vs. safe decisions, independent of which specific visual element was being compared? This folds what was RQ1 into RQ2's broader territory rather than answering the original AOI-anchored wording. The rescoping was confirmed still necessary after a dedicated, bounded spike specifically looking for a cheaper AOI-recovery path (the codebase's own aoi_mapper.py) came back negative - not an assumption carried over unchecked from an earlier pass. The AOI-anchored version of RQ1 remains a legitimate deliverable, just for new data collected with head-pose logging (Objective 3), not the existing 140 sessions.

Phased, not deferred - EEG now has a real starting n. The gaze+behaviour paper is still the first, primary analysis (full 140-participant coverage). A second, EEG-extended analysis can proceed in parallel on the 67 participants (47.9%) with decoded, session-linked, quality-screened EEG.

Complete research architecture



Unchanged pipeline diagram from the prior revision - still accurate, with one caveat this revision adds: the "Feature extraction" box's AOI features are deliverable only for new data collected with head-pose logging (Objective 3); on the existing 140 sessions, the AOI-free cluster-shape features described above stand in for them.

Experimental plan

Design: within-subjects - each participant already sees a mix of safe and attack kiosks across ~20 stations per session. The automation-bias sub-study (correct/absent/malicious AI advice) should also be within-subject, randomized and counterbalanced across stations.

Sample size reasoning: n=140 (verified on disk) is large by the standards of comparable gaze/EEG human-factors studies. EEG is decoded and usable for 67 of 140 (47.9%) - budget any EEG-involving analysis for n=67, not 140, and treat it as a secondary track.

Practical constraint for the automation-bias sub-study specifically, new this revision: the Gemini API's free-tier quota is exactly 20 requests/day, per project, per model - a single reliability-testing batch of 20 trials consumes a full day's quota by itself. Budget multi-day testing windows, not a single session, for finishing the compliance-rate estimate or iterating on prompt wording.

Statistics (primary, confirmatory): generalized linear mixed-effects models, e.g. decision_outcome ~ attack_type * AI_recommendation_validity + (1|participant) + (1|station). Effect sizes with confidence intervals, not p-values alone; FDR correction across the many gaze features tested.

What would count as failure, stated up front: if gaze features show no reliable difference between risky and safe decisions after FDR correction, or if a trained model cannot beat the behaviour-only baseline on held-out participants, that is a reportable negative result, not a reason to keep searching for significance.

Implementation roadmap

Quick wins (days, minimal risk)

- Lower Shadows::textureScalar and raise Shadows::disable; measure FPS delta before/after.
- Lower Reflect::refractTexScale - already applied; needs measuring.
- Set the 2D mirror view to "Empty" and confirm OpenXR debug mode is off before any FPS run.
- Reconcile the two divergent main.py/config copies (outer backend/ vs. deployed EVChargingGame/backend/) - a reproducibility risk for cross-date session comparisons.

Medium (1-2 weeks)

- Run a proper CPU/GPU/physics profiling pass to replace the UNVERIFIED engine-level guesses in Objective 1 with measured numbers.
- EEG decode-and-QC done (76/76 decoded, 67/76 usable). Remaining: ask whoever ran participant 058's session whether its real data exists under a different folder/ID.
- Build the gaze-only + behaviour-only feature extraction and baseline models on the existing 140-session dataset.
- **Done this revision:** the correct/absent/malicious condition variable is implemented in ai_assistant.py, tested against the live model (baseline wording 16/20 = 80% compliance, 95% CI [58.4%, 91.9%]; stronger wording still short of a full batch at 6/6 = 100% due to API quota limits). **Remaining:** finish the stronger-wording batch once quota resets and decide, with the mentor, between further prompt-engineering vs. a deterministic override. Fix the RAG store's cross-session contamination before using these results in any

recommendation-quality analysis - now confirmed to affect 100% of the 38 sessions that used the assistant, with an exclusion list ready at backend/rag_contaminated_sessions.json.

- **Updated this revision:** AOI mapping is now conclusively blocked on the existing 140 sessions via two independent, exhausted routes (world-space: no head pose logged; the codebase's own 2D aoi_mapper.py, falsified against ground truth and found to have a vergence/depth issue, not a fixable 2D calibration bug) - not a decision pending, a closed door for existing data. The AOI-free fallback covers a rescoped version of RQ1 (see Experimental plan). Going forward: the head-pose-logging fix is scoped but blocked on locating the external gaze-capture application and the flowgraph scenario file; a cross-process timestamp-sync design is proposed but not implemented.

Deep (architectural, next phase)

- Full multimodal pipeline (gaze + EEG + behaviour) once EEG is verifiably collected and synchronized for a meaningful subset of new participants.
- AI-warning integration that actually consumes the trained risk model's output, with the correct/absent/malicious recommendation conditions for the automation-bias sub-study.
- New data collection round incorporating the Objective 3 schema, head-pose logging, and QC checks above.

Professor presentation summary

A one-page version of this audit's load-bearing findings, for the next meeting with Prof. Dutt and Ranik Goyal.

- **1.** The dataset is stronger than the mid-sem report's Table 6 suggests on gaze, and EEG is now a real, decoded, usable dataset for about half the participants. All 140 sessions contain complete Varjo gaze data; 67 of 140 participants (47.9%) have decoded, session-linked, quality-screened EEG.
- **2.** The in-sim "AI assistant" already is the kiosk-trust recommendation system the script and report describe. It's image-based rather than gaze/EEG-based (that part was right), but it is the same system, not a different one.
- **3.** FPS optimization is bounded by what's inspectable, and still blocked on a Windows measurement that hasn't happened yet. The 120 FPS target should be revised to a stable 90, matching the XR-4's refresh rate. This is the one item on this list that only you can unblock.
- **4. (New) The automation-bias manipulation is no longer just a plan - it's built and empirically tested.** A condition flag (correct/absent/malicious) was added to the existing AI assistant with no behavior change by default, and tested against the live Gemini model: baseline wording complied with the malicious override in 16 of 20 real trials (80%); a stronger imperative wording complied in all 6 valid trials run so far (100%, but too few to call it reliably better than the baseline yet). Testing is currently paused by a hard 20-request/day API quota, not a design problem - expect this number to firm up once that resets and the remaining trials run. Separately, the AI assistant's shared memory (RAG store) is now confirmed to have affected literally all 38 of the 140 sessions that used it, not just "a shared store exists" - every assistant interaction in the dataset carries this caveat until the store is reset.
- **5. (New) AOI-level gaze analysis ("did they look at the coupon or the price") is a closed question for the existing 140 sessions, confirmed twice over, not once.** Beyond the world-space blocker (gaze has no head pose to anchor it to), a second candidate fix already sitting in the codebase (aoi_mapper.py, a 2D screen-space approach that doesn't need head pose) was tested against real participants' actual kiosk choices and found to assign gaze to the *wrong* kiosk with high confidence, consistently - and a bounded effort to repair it surfaced a deeper issue (eye-vergence/depth dependence) that rules out a simple fix. The gaze-behaviour-shape fallback (unlabeled fixation clustering) still works and should anchor the primary analysis instead; true AOI labels are a new-data-collection deliverable once head-pose logging lands, pending locating the separate gaze-capture tool.

The literature review and gap analysis (Objective 4) remain the project's strongest asset as-is - no changes recommended beyond verifying the four citations it already flags [to verify].